

Statement of Work for Analyzing Patents

1. Background

NASA is looking for transformative concepts in aeronautics—concepts that could dramatically improve aircraft speed, fuel efficiency, safety and (or) enable opening new markets and new services from the aviation industry. Our researchers are actively engaged in innovating and testing these concepts. One item that could help them innovate are analyses of immense knowledge buried within the US patents. Perhaps these analyses will point to knowledge from previous patents which could not be implemented but can now be implemented and yield multiple transformative concepts. Perhaps a unique combination of knowledge and patents could provide another innovation. We are seeking these analyses.

2. Scope

Effort is to extract scientific and engineering knowledge from US aviation-related patents since 1976 and assess if, and how, this knowledge could be repurposed for transformative concepts.

3. Objectives

Effort is to identify actionable insights for transformative concepts by analyzing the knowledge used to create the patents and the patents themselves by providing:

1. **Clear, data-driven understanding** of historical and current technological innovation in aviation through comprehensive patent analysis spanning 50 years (1976–present)
2. **Overview of aviation technology applications** and real-world utility of patents in practical contexts
3. **Identification of overlooked connections** between technology domains within aviation to uncover synergies yielding economic impact, competitive advantage, and strategic opportunity

4. Tasks

A. Patent Data Extraction

Systematically collect and compile all U.S. aviation-related patents filed since 1976 to present from authoritative sources including the United States Patent and Trademark Office (USPTO), Google Patents, and public databases. This task includes:

- Building a comprehensive data pipeline to extract all relevant aviation patents
- Ensuring complete metadata capture including filing dates, assignees, patent abstracts, claims, and classifications (CPC/IPCR)
- Filtering and organizing data for accuracy and relevance

B. Patent Categorization

Organize extracted patents into functional taxonomies reflecting real-world application and purpose, including but not limited to:

- Airframe and structural systems (morphing wings, structural health monitoring)
- Propulsion systems (hybrid-electric architectures, rotating detonation engines)
- Avionics and safety systems
- Sustainability and efficiency innovations

C. Patent Combination Analyses

Apply AI and data science methodologies to:

- Identify patent clusters and thematic groups
- Explore cross-cluster synergies and connections between unrelated technology domains
- Map high-potential patent combinations with strategic summaries and use case examples
- Deliver visual maps and structured exploration of breakthrough innovation opportunities

5. Deliverables

1. **Excel Dataset** – Comprehensive, structured compilation of extracted, categorized, and analyzed patent data
2. **Analysis Summary** – Strategic analysis identifying technology trends, emerging themes, and innovation opportunities within the aviation patent landscape
3. **Visual Maps** – Graphical representations of patent combinations, cross-technology synergies, and opportunity mapping

Phase	Week	Deliverable
Kickoff & Scope Definition	Week 1	Project initiation; data source confirmation
Patent Data Extraction	Weeks 1–3	Complete pipeline build and data compilation
Patent Categorization	Weeks 2–4	Taxonomy development and AI-guided classification
Patent Combination Analysis	Weeks 3–5	Synergy identification and opportunity mapping
Quality Control & Review	Week 5–6	Validation, final dataset assembly, client review
Delivery	Week 8	Final Excel dataset and analysis reports

6. Travel Requirements: None.

7. References: More details about our aeronautics related work is found here:
<https://www.nasa.gov/directorates/armd/>

8. Government Furnished Property and Information: N/A

9. Contract type: Firm-Fixed Price

10. Place of Performance: Contractor facility

11. Period of Performance: Eight weeks from date of award